High Altitude Balloons

Launching high-altitude research devices to 100,000ft!

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HAB LAUNCH

Goal: Successfully launch and recover a balloon with camera footage.

How it works: Helium lifts the balloon to a desired float altitude. As the balloon rises, the gas inside expands and eventually pops, releasing a parachute as the payload falls to earth.

Payload: An insta360 camera, power bank, GPS tracker and a mini Roaree mounted to a wooden triangle frame.





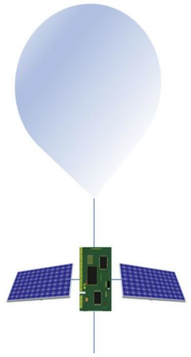


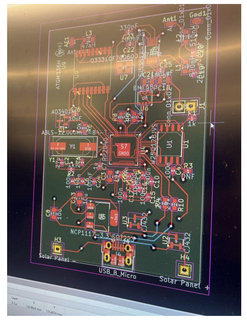
PICO BALLOONS

How it works:

- Pico balloons are small, lightweight Mylar balloons designed for long-duration flights.
- Carry a basic payload of electronics, radio transmitter, solar cells, and helium
- Capable of circumnavigating the globe collecting data on temperature and wind patterns.







Launch Mechanics:

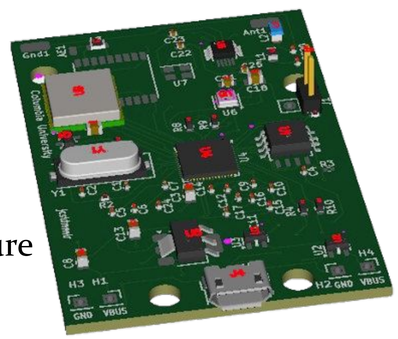
Investigated and organized methods for

- Stretching, filling, and sealing balloon
- Attaching the PCB and antenna (30ft)
- Calculating helium amount

Printed Circuit Board (PCB):

Developed a custom combination flight computer, integrating:

- Combination humidity, pressure, and temperature sensor (Bosch)
- GPS receiver
- WSPR Radio transmitter – HF radio
- RP2040 microcontroller

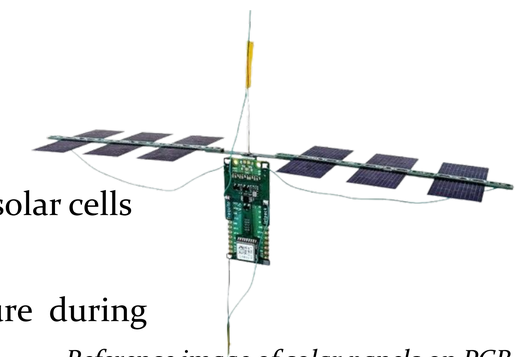


Custom PCB in KiCAD

Power:

The flight computer is powered by 6 micro solar cells

- No battery
- Only transmits position and temperature during daylight hours



Reference image of solar panels on PCB

RESULTS AND FUTURE PLANS

Accomplishments:

Successful launch to 100,000 ft, and recovery of footage!

The payload landed in the tree of a private homeowner, who kindly recovered the entire payload for us. We were able to recover footage from the launch from Delhi, NY.

Successful development and manufacturing of custom PCB!

The custom PCB integrates all electrical components required for a successful Pico balloon lifespan.

Next steps:

Launch a fleet of 10 Pico balloons from Morningside Heights and collect data from around the world!

LEARN MORE

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